

Regime-Conditional Reward-Free Exploration: Learning Recurrent Dynamics and Diagnosing Deployment Modes

Anonymous Authors

Abstract

Reward-free exploration (RFE) asks a learner to collect data without observing rewards, then to produce a near-optimal policy for an arbitrary reward revealed afterwards. Standard RFE theory assumes a single stationary transition kernel. We study a different obligation that arises when dynamics themselves cycle through unknown operating regimes. During reward-free pretraining the environment follows a hidden piecewise-stationary sequence of recurring kernels. At deployment the learner receives a short reward-free diagnostic phase in one previously seen regime, after which an arbitrary reward is revealed and no further interaction is allowed. The resulting problem, *regime-conditional RFE*, jointly requires (i) contamination-safe demultiplexing of an unsegmented stream, (ii) reward-uniform model learning for every recurring regime, and (iii) reward-independent identification of the active deployment kernel.

We introduce Detect–Match–Explore (DME), a modular construction that probes transitions, quarantines uncertain boundaries, matches returning regimes, aggregates regime-specific samples, and diagnoses deployment before planning. We prove: a finite-horizon simulation lemma; a diagnostic KL chain rule; inconsistency of pooled estimators when action semantics conflict; a boundary contamination identity that motivates quarantine; uniqueness of confidence-set matching under a separation margin; a recurrence-versus-restart sample-accounting lemma; a conditional PAC reduction and error decomposition; known-model diagnosis rates; a simplified i.i.d. probe-stream two-window test; SEG for probe-window DME under an explicit window length (Theorem 6); a robust-diagnosis reduction when confidence sets have a positive TV margin; zero-observation and finite-diagnostic impossibilities; and occupancy and behavioral-equivalence obstructions. We do not prove an M -mode direct-sum minimax lower bound, and we do not claim SEG for the short-window residual heuristic used in the short-dwell benchmark.

A checked ten-seed finite-horizon tabular pilot shows that pooling incompatible regimes is costly (mean worst-reward gap 0.135), while recurrence-aware learning attains gap 0.00138 and saves 48.1 transitions versus restarting. The paired advantage over restart is only 0.00110 (bootstrap 95% interval [0.00057, 0.00178]), below a predeclared 0.01 practical margin. On RFE-Recurrent-Bench the corrected detector yields paired improvement 0.073 ([0.040, 0.109]) and the suite gate passes; MiniGrid still fails because pooling remains better. The paper therefore supplies a precise objective, algorithms, modular theory, and a passing short-dwell benchmark; it does not claim SEG for that short-window heuristic, nor an M -fold direct-sum bound.

1 Introduction

Reward-free exploration separates data collection from task specification [14, 15, 28, 19]. A learner interacts with an unknown Markov decision process (MDP) without observing rewards and must later, given an arbitrary reward function from a declared class, output a near-optimal policy without collecting more data. This is stronger than online reinforcement learning for a single known reward [24, 13]: the same dataset must support every subsequently specified objective.

Existing RFE guarantees assume one stationary transition kernel. Many systems instead revisit a small family of dynamics. A warehouse robot alternates payloads; a network returns to traffic

modes; a simulator cycles physical parameters. Pooling incompatible regimes biases the estimated kernel. Restarting after every change wastes samples when a regime returns. Even a perfect library of models is unusable at test time if the learner does not know which dynamics are currently active.

Non-stationary reinforcement learning does not resolve this objective. Windowing, discounting, and restart methods optimize dynamic regret while rewards are observed [6, 26]. Detection wrappers such as DARLING probe transitions and rewards to restart a stationary online learner [10]. Recurring-context methods detect changes and reuse reward-specific policies [1]. Latent and contextual MDPs treat hidden identity as part of a reward-bearing control problem [11, 16]. Active MDP detection designs controls that distinguish a *known* candidate family [9]. None of these lines simultaneously (a) learns the candidate family from an unsegmented *reward-free* stream, (b) preserves those data for *uniform* post-hoc planning, and (c) identifies the deployment kernel *before* the reward is revealed.

Problem in one sentence. Learn, without rewards, a library of transition models for a recurring hidden family, then use a short reward-free diagnostic interaction to select the active model, then plan for an arbitrary subsequently revealed reward.

Why the problem is not a trivial composition. Running stationary RFE independently in each detected segment would ignore recurrence. Matching segments without quarantining mixed boundary data can contaminate every future occurrence of a mode. Diagnosing the deployment mode after the reward is revealed is a weaker problem, because the diagnostic policy may then specialize to that reward. Diagnosing with *no* deployment interaction is impossible in general, even with an exact library (Theorem 7). The interesting object is therefore the joint contract: valid demultiplexing, reward-uniform reuse, and reward-independent diagnosis.

Contributions.

1. **Objective.** We define regime-conditional RFE, including reward normalization, a uniform PAC criterion, a controlled diagnostic information coefficient, and an assumptions ledger with necessity witnesses.
2. **Algorithm.** We specify Detect–Match–Explore (DME) through explicit module interfaces: predictable probing, change detection, boundary quarantine, permutation-invariant recurrence matching, a stationary RFE routine, robust deployment diagnosis, and abstention.
3. **Theory.** We prove a simulation lemma and KL chain rule (Lemmas 1–2); pooling inconsistency and boundary contamination (Theorems 1–2); matching uniqueness and recurrence accounting (Lemmas 4–5); modular PAC and error decomposition (Theorems 3–4); known-model diagnosis rates (Proposition 1 and Theorem 5); a probe-stream two-window test (Lemma 6); SEG for probe-window DME (Theorem 6); robust diagnosis under a TV margin (Proposition 2); impossibilities (Theorems 7–10); and a sample-complexity corollary under SEG (Corollary 1). The short-window residual heuristic used in the short-dwell benchmark is not covered by Theorem 6. An M -fold direct-sum lower bound remains open.
4. **Evidence.** We report a saturated long-dwell tabular pilot whose practical gate fails, a MiniGrid stress test whose pooling gate fails, and RFE-Recurrent-Bench, whose short-dwell gate passes. Failures are retained. A **stress** profile with occupancy imbalance and $M \in \{2, 3, 5\}$ is implemented; it is not a replacement for the retained hashes.

2 Related work

Stationary reward-free exploration. Jin et al. [14] formalize tabular RFE and prove that a dataset sufficient for accurate transition estimation yields ε -optimal policies for every bounded reward, together with a single-mode lower bound of order $\Omega(S^2AH^2/\varepsilon^2)$ under their stated conventions. Adaptive RFE allocates exploration effort using estimated model uncertainty [15]. Subsequent work approaches minimax rates and clarifies horizon dependence for time-homogeneous versus time-inhomogeneous kernels [28, 19]. These results supply candidate *per-regime* modules. They do not address hidden segmentation. Detector-filtered trajectories cannot inherit a stationary theorem unless the sampling and stopping-time contract of that theorem is verified.

Optimism and model-based online RL. UCRL2 and related confidence-set methods plan in optimistic models constructed from visit counts [13, 2, 8]. They observe rewards online and optimize a single objective. Our stationary module may use the same concentration tools, but the output must be a planner that remains valid for every later reward. Naming a count-bonus explorer “UCRL-RFE” is therefore incorrect unless the published algorithm is actually implemented.

Non-stationary RL. When rewards and transitions drift, dynamic regret is the usual criterion. Variation-budget analyses restart or inflate bonuses as a function of total variation [6, 21]. Black-box reductions such as MASTER convert stationary algorithms into non-stationary procedures via tests and restarts [26]. DARLING separates detection from learning by inserting probe episodes and monitoring reward and transition streams directly, obtaining near-minimax dynamic regret in piecewise-stationary tabular and linear MDPs under reachability and dwell conditions [10]. These methods observe the training reward and need not produce a reusable reward-uniform library. Recurrence is typically treated as a new stationary segment rather than as an opportunity to pool data.

Recurring contexts, latent MDPs, and hidden modes. Contextual MDPs attach a hidden context to an otherwise ordinary MDP [11]. Latent-MDP analyses give regret and lower bounds when the unobserved model is redrawn, often once per episode, under statistical gap assumptions [16]. MBCD detects changepoints and reuses previously learned models or policies when a context returns [1]. These works establish that hidden identity is not itself a new primitive. Our additional burden is contamination-safe demultiplexing for *arbitrary future rewards*, together with diagnosis *before* the reward is known.

Active identification of known MDPs. Duan et al. [9] synthesize policies that distinguish a fixed known family. Our deployment phase reduces to a robust version of that problem after models have been estimated. Training is different: the family, boundaries, and recurring assignments are unknown. This distinction motivates confidence-set diagnosis and explicit rejection when models overlap.

Sequential testing and change detection. Time-uniform confidence sequences provide valid inference under continuous monitoring [12]. Classical inspection schemes [23, 18] control delay for i.i.d. streams; they do not by themselves yield an MDP segmentation contract. Bretagnolle–Huber and related inequalities convert KL divergence into testing error [3, 17, 5, 25]. Weissman et al. give L_1 deviation bounds for empirical distributions [27]. We use these tools for diagnosis lower bounds, for a simplified probe-stream lemma, and for Theorem 6.

Table 1: Positioning. “Library” means a reward-uniform transition model usable for arbitrary post-hoc rewards. “Diagnosis” means identifying the active kernel *before* the reward is revealed.

Method	Reward-free training	Recurrence reuse	Reward-uniform library	Pre-reward diagnosis	Learns hidden family
Stationary RFE [14]	yes	n/a	yes	n/a	n/a
UCRL2 / optimistic RL [13]	no	n/a	no	n/a	n/a
MASTER / variation budget [26, 6]	no	no	no	n/a	no
DARLING [10]	no	typically no	no	n/a	no
MBCD [1]	no	policies	no	no	yes
Latent MDP [16]	no	n/a	no	n/a	yes
MDP detection [9]	n/a	n/a	n/a	yes	no
DME (this paper)	yes	yes	yes	yes	yes

Multi-task, latent, and block structure. Multi-task RL analyses sample sharing across related reward tasks in a stationary environment [4]. Block-MDP and rich observation models address latent state, not recurring transition regimes [20]. Surveys of non-stationary RL collect the broader empirical and theoretical landscape [22].

The last row is an *objective* claim: DME is designed for that intersection. It is not a claim that the implementation meets the corresponding PAC guarantee.

3 Setting

Let $\mathcal{S}$ and $\mathcal{A}$ be finite, $S = |\mathcal{S}|$, $A = |\mathcal{A}|$, and $H \geq 1$ the episode horizon. The initial-state distribution ρ is known and common to all modes. An episodic kernel is $P = (P_h)_{h=1}^H$ with $P_h(\cdot | s, a) \in \Delta(\mathcal{S})$. There is an unknown family

$$\mathcal{P}_\star = \{P^{(1)}, \dots, P^{(M_\star)}\}, \quad M_\star \leq \bar{M},$$

where $\bar{M}$ is known. Distinct labels mean distinct kernels; permutations are unidentifiable and have no meaning.

Training. Training lasts K complete episodes. Before episode $k \in [K]$, nature selects a hidden label $z_k \in [M_\star]$; the entire episode is generated by $P^{(z_k)}$. Switches occur only between episodes. The learner observes $(s_{k,h}, a_{k,h}, s_{k,h+1})_{h=1}^H$ but neither z_k , switch boundaries, nor rewards. Actions may depend on the full observation history and private randomness. The schedule is piecewise constant: if its maximal constant blocks are $[b_j, e_j]$, the minimum dwell condition is $e_j - b_j + 1 \geq D_{\min}$. Modes may recur in any order. Guarantees are conditional on every deterministic schedule satisfying the assumptions below; there is no implicit i.i.d. mode prior.

Deployment. Nature then selects a deployment label $Z \in [M_\star]$ that appeared during training and for which the useful-sample quota below is met. Z remains fixed through q fresh reward-free diagnostic episodes and subsequent execution. Diagnostic policies may be history-dependent but observe no rewards. After the q -th diagnostic episode an arbitrary reward

$$r = (r_h)_{h=1}^H, \quad r_h : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1], \quad \sup_\tau \sum_{h=1}^H r_h(s_h, a_h) \leq 1 \quad (1)$$

holds. We refer to this pathwise bound as condition (R). The function r is revealed in full. There is no post-reward interaction. The learner returns a (possibly nonstationary Markov) policy $\hat{\pi}_r$.

Write

$$V_P^\pi(r) = \mathbb{E}_{\rho, P, \pi} \left[\sum_{h=1}^H r_h(s_h, a_h) \right], \quad V_P^\star(r) = \sup_{\pi} V_P^\pi(r).$$

Pathwise normalization (1) keeps every value and gap in $[0, 1]$. Replacing it by $r_h \in [0, 1]$ without a pathwise bound multiplies every unit loss by at most H .

Definition 1 (Regime-conditional PAC-RFE). *An algorithm is (ε, δ) -PAC if, for every admissible family, schedule, and eligible deployment mode,*

$$\Pr \left\{ \sup_{r \in \mathcal{R}_1} [V_{P(z)}^\star(r) - V_{P(z)}^{\hat{\pi}_r}(r)] \leq \varepsilon \right\} \geq 1 - \delta. \quad (2)$$

The supremum lies *inside* the event: one learned library must support every post-hoc reward simultaneously. Probability covers training, diagnostics, and algorithmic randomness. Unseen deployment modes may return REJECT; they receive no near-optimality guarantee.

3.1 Controlled diagnostic information

A q -episode diagnostic strategy σ maps observed histories to action distributions. Let $\mathbb{P}_m^\sigma$ be the law of the complete q -episode transcript under $P^{(m)}$. Hellinger affinity is $\text{Aff}(P, Q) = \int \sqrt{dP dQ}$. For a known finite family define

$$\mathcal{B}_q(\mathcal{P}_\star) = \sup_{\sigma} \min_{m \neq \ell} [-\log \text{Aff}(\mathbb{P}_m^\sigma, \mathbb{P}_\ell^\sigma)]. \quad (3)$$

with $-\log 0 = +\infty$. This coefficient includes reachability: differences at state-actions never visited by σ contribute nothing. For lower bounds we use $\mathcal{K}_q^\sigma(m, \ell) = D_{\text{KL}}(\mathbb{P}_m^\sigma \| \mathbb{P}_\ell^\sigma)$. With estimated models the robust design objective is

$$\hat{\sigma} \in \arg \max_{\sigma} \min_{m \neq \ell} \inf_{P \in \mathcal{C}_m, Q \in \mathcal{C}_\ell} -\log \text{Aff}(\mathbb{P}_P^\sigma, \mathbb{P}_Q^\sigma). \quad (4)$$

If the inner infimum is zero, a certified implementation must abstain.

3.2 Assumptions

Assumption 1 (Probe reachability). *There is a finite probe set $\mathcal{X} \subseteq [H] \times \mathcal{S} \times \mathcal{A}$ such that for every $x = (h, s, a) \in \mathcal{X}$ some reward-free policy reaches (s, a) at step h with probability at least $\mu_{\text{probe}} > 0$ in every mode.*

Assumption 2 (Controlled separation). *Behaviorally distinct modes satisfy $\mathcal{B}_q(\mathcal{P}_\star) > 0$. Quantitative diagnosis at level δ_{id} requires a margin such as $\mathcal{B}_q(\mathcal{P}_\star) \geq \log((\bar{M} - 1)/(2\delta_{\text{id}}))$.*

Assumption 3 (Useful occupancy). *After quarantine, every deployment-eligible mode m contributes at least $N_{\text{RFE}}^{(m)}(\eta, \delta_{\text{rfe}})$ accepted episodes that satisfy the input contract of the chosen stationary RFE routine.*

Assumption 4 (Dwell). *$D_{\min} \geq 2B + L_{\text{probe}} + L_{\text{match}} + L_{\text{useful}}$, where B is the per-end quarantine budget and the three L terms are declared disjoint episode budgets unless a module theorem proves valid reuse.*

Assumption 5 (Fixed seen deployment; transition-only modes). *The deployment kernel is one training mode and remains fixed. Modes differ only through $P^{(m)}$. Rewards are absent until after diagnosis.*

Assumption 6 (Adaptive validity). *Every confidence set $\mathcal{C}_m$ is time-uniform, or is formed at a declared stopping time with a valid adaptive-data argument [12].*

Necessity witnesses. If two kernels differ only at unreachable triples, all transcript laws coincide (Theorem 10). A mode that appears for one episode cannot in general be learned to arbitrary accuracy (Theorem 9). If modes alternate every episode while detection needs a pure window of length two, no window detector can succeed. Reward-only “goal drift” leaves the reward-free law unchanged and is undetectable. Fixed-sample multinomial intervals applied after a data-dependent alarm can undercover and permanently contaminate a library.

3.3 Segmentation contract

A segmentation module returns an accepted set $U \subseteq [K]$, a quarantine $Q = [K] \setminus U$, an assignment $c : U \rightarrow [\widehat{M}]$, and confidence sets $(\mathcal{C}_i)_{i=1}^{\widehat{M}}$.

Definition 2 (SEG). *The output satisfies $\text{SEG}(B, \mathbf{n}, \delta_{\text{seg}})$ if, with probability at least $1 - \delta_{\text{seg}}$, there is a bijection $\phi : [\widehat{M}] \rightarrow [M_\star]$ such that simultaneously: (i) purity: $c(k) = i$ implies $z_k = \phi(i)$; (ii) localization: $Q \cap [b_j, e_j]$ is contained in the first and last B episodes of that block; (iii) no split or merge: all accepted episodes of true mode m , including nonadjacent recurrences, receive $\phi^{-1}(m)$; (iv) quota: each eligible mode meets its RFE input contract on accepted trajectories; (v) confidence: $P^{(m)} \in \mathcal{C}_{\phi^{-1}(m)}$ jointly despite adaptive alarms.*

SEG is an analysis event involving hidden truth, not a runtime flag. Theorem 6 establishes SEG for probe-window DME. The short-window residual heuristic is not that algorithm.

4 Detect–Match–Explore

DME is specified by module interfaces. The empirical artifact instantiates those interfaces; it is not a proof that they hold.

Algorithm 1 (Detect–Match–Explore).

Input: budgets $K, q, \bar{M}$; probe family; modules DETECT, MATCH, RFE, DESIGN.

Initialize library $L = \emptyset$ and buffer W .

Training. For episode $k = 1, \dots, K$:

1. Execute a predictable probe or the policy requested by the active RFE instance. Observe transitions; append the trajectory to W .
2. Update DETECT from the predictable policy and observed transitions only.
3. On alarm: quarantine the declared pre/post windows; close the accepted interior of W ; form a time-valid confidence set $\mathcal{C}_{\text{new}}$.
4. Set $i = \text{MATCH}(\mathcal{C}_{\text{new}}, L)$. Pool only a certified unique recurrence; otherwise create a new entry. Ambiguity triggers more probes or abstention; never silent pooling.
5. Resume the matched mode’s RFE instance on accepted data.

Close the final block analogously. If a module certificate fails, return ABSTAIN.

Deployment. Set $\hat{\sigma} = \text{DESIGN}(\{\mathcal{C}_i\}, q)$ via (4). If certified pairwise information is zero, abstain. Execute $\hat{\sigma}$ for q episodes. Identify $\hat{m}$ by a declared robust test. If no label is certified, return REJECT/UNSEEN. After r is revealed, return $\Pi_{\hat{m}}(r)$.

Stationary RFE module. An $\text{RFE}(\eta, \delta_{\text{rfe}}, N)$ module maps N pure admissible episodes from one unknown P to a planner Π such that

$$\Pr\left\{\sup_{r \in \mathcal{R}_1} [V_P^*(r) - V_P^{\Pi(r)}(r)] \leq \eta\right\} \geq 1 - \delta_{\text{rfe}}.$$

Any invocation must state whether the kernel class is time-homogeneous and must match the reward scale of the cited result. Detector-generated data are not automatically valid input.

Implementation used in experiments. The code uses a conditional two-window detector on pre-update predictive residuals, total-variation matching of segment models, quarantined count tables, finite-horizon value iteration on valid states, and prefix-only deployment matching. Ground-truth labels are confined to environment generation, evaluation, and methods whose names begin with **oracle**. The collector is a least-visited-action optimistic baseline, not a named minimax RFE algorithm.

Matching module.

Algorithm 2 (Confidence-set matching).

Input: new set $\mathcal{C}_{\text{new}}$; library $(\mathcal{C}_i)_{i \in L}$; margin $\gamma > 0$.
For each $i \in L$ set $d_i = \sup_{h,s,a} \inf_{P \in \mathcal{C}_{\text{new}}, Q \in \mathcal{C}_i} \text{TV}(P_h(\cdot | s, a), Q_h(\cdot | s, a))$.
Let $I_0 = \{i : d_i = 0\}$ and $I_\gamma = \{i : d_i \leq \gamma\}$.
If $|I_0| = 1$ and $I_\gamma = I_0$, return that index (certified recurrence).
If $I_0 = \emptyset$ and $\min_i d_i > \gamma$, return NEW.
Otherwise return ABSTAIN (overlap or multiple inclusions).

Lemma 4 shows that a separation margin larger than twice the confidence radius makes the certified-recurrence branch unique. Silent pooling when d_i is only small, not zero, is not licensed by that lemma.

Diagnosis module.

Algorithm 3 (Robust deployment diagnosis).

Input: confidence sets $(\mathcal{C}_i)$; budget q ; error δ_{id} .
If the inner infimum in (4) is zero for every σ , return ABSTAIN.
Otherwise execute a certified $\hat{\sigma}$ for q episodes.
Compute pairwise robust affinities on the observed transcript.
If a unique index i has all competitors rejected at level δ_{id} , return i ; else return REJECT/UNSEEN.

Proposition 1 and Theorem 5 analyse the known-model special case $\mathcal{C}_i = \{P^{(i)}\}$. Proposition 2 extends the same test when each $\mathcal{C}_i$ is a TV ball of radius $\rho < \Delta/4$.

5 Theory

We first record analytic facts, then structural reasons pooling and unquarantined boundaries fail, then the modular reduction, then information-theoretic obstructions. Complete proofs are in Appendix B. Throughout, $\text{TV}(P, Q) = \frac{1}{2} \|P - Q\|_1$.

5.1 Value and information identities

Lemma 1 (Finite-horizon simulation under (1)). *Let $P, \hat{P}$ be episodic kernels and π any policy. If $\text{TV}(P_h(\cdot | s, a), \hat{P}_h(\cdot | s, a)) \leq \xi$ for every (h, s, a) in the support of π under both kernels, then $|V_P^\pi(r) - V_{\hat{P}}^\pi(r)| \leq H\xi$ for every $r \in \mathcal{R}_1$. Consequently, if $\hat{\pi}$ is optimal in $\hat{P}$,*

$$V_P^*(r) - V_{\hat{P}}^{\hat{\pi}}(r) \leq 2H\xi.$$

Lemma 2 (Diagnostic KL chain rule). *For any adaptive σ ,*

$$\mathcal{K}_q^\sigma(m, \ell) = \sum_{t=1}^q \sum_{h=1}^H \mathbb{E}_m [D_{\text{KL}}(P_h^{(m)}(\cdot | s_{t,h}, a_{t,h}) \| P_h^{(\ell)}(\cdot | s_{t,h}, a_{t,h}))].$$

State-action-stage triples never reached contribute zero. If some reached triple has $P_h^{(\ell)}(\cdot | s, a)$ not absolutely continuous w.r.t. $P_h^{(m)}$, the KL is infinite.

Lemma 3 (Affinity, Pinsker, and Le Cam). *For any distributions P, Q on a common space,*

$$2 \text{TV}(P, Q)^2 \leq D_{\text{KL}}(P \| Q), \quad \text{TV}(P, Q) \leq \sqrt{1 - \text{Aff}(P, Q)^2}.$$

Consequently $\text{Aff}(P, Q) \leq \sqrt{1 - \text{TV}(P, Q)^2} \leq \exp(-\frac{1}{2} \text{TV}(P, Q)^2)$.

5.2 Why pooling and unquarantined mixing fail

Theorem 1 (Inconsistency of pooled kernels). *Let $P \neq Q$ and $\alpha \in (0, 1)$. Write $M = \alpha P + (1 - \alpha)Q$ for the mixture kernel acting coordinatewise. Then for every (h, s, a) ,*

$$\text{TV}(M_h(\cdot | s, a), P_h(\cdot | s, a)) = (1 - \alpha) \text{TV}(Q_h(\cdot | s, a), P_h(\cdot | s, a)).$$

Consequently there exist two-mode instances, a mixing weight $\alpha = 1/2$, and a reward $r \in \mathcal{R}_1$ such that every policy that is optimal for M has gap $1/2$ in both true modes. In particular, the pooled MLE cannot be uniformly near-optimal over $\mathcal{R}_1$ even with infinite mixed data.

Theorem 2 (Boundary contamination). *Let a window of n episodes contain n_0 trajectories from P and $n_1 = n - n_0$ from Q , and let $\widehat{M}$ be the empirical kernel on that window. Then*

$$\mathbb{E}[\widehat{M}_h(\cdot | s, a)] = \frac{n_0}{n} P_h(\cdot | s, a) + \frac{n_1}{n} Q_h(\cdot | s, a)$$

at every (h, s, a) that is sampled independently of the mode label given the policy. If this mixed window is accepted into a library entry for P , the expected TV bias equals $(n_1/n) \text{TV}(Q_h, P_h)$. Quarantine of length B removes the bias whenever every mixed window is contained in the discarded $2B$ episodes at block ends.

Theorem 2 is an identity, not a detector guarantee: it says what quarantine *must* accomplish, not that any particular alarm does so.

Lemma 4 (Unique certified match). *Suppose $\min_{m \neq \ell} \max_{h,s,a} \text{TV}(P_h^{(m)}, P_h^{(\ell)}) \geq \Delta$ and every confidence set $\mathcal{C}_i$ has TV radius at most $\rho < \Delta/4$ around its true kernel. Then Algorithm 2 with $\gamma = \Delta/2$ returns the unique true recurrence when the new set is a recurrence, and returns NEW when the new kernel is an unseen mode at the same separation.*

Lemma 5 (Recurrence versus restart accounting). *Suppose SEG holds with quarantine B per block end. If mode m occupies J blocks of length at least $D_{\min}$, restart that discards previous blocks retains at most $D_{\min}$ useful episodes, while recurrence matching retains at least $J(D_{\min} - 2B)$ accepted episodes. If a stationary RFE module requires N episodes for accuracy η , restart fails the module contract whenever $D_{\min} < N \leq J(D_{\min} - 2B)$.*

5.3 Modular reduction

Theorem 3 (Conditional modular PAC reduction). *Suppose SEG fails with probability at most δ_{seg} . Conditional on SEG, each of the $M_\star$ stationary RFE calls has reward-uniform error at most η and failure probability at most δ_{rfe} . Conditional on those events, deployment identification fails with probability at most δ_{id} . Then*

$$\Pr\left\{\sup_{r \in \mathcal{R}_1} \text{gap}(r) \leq \eta\right\} \geq 1 - \delta_{\text{seg}} - M_\star \delta_{\text{rfe}} - \delta_{\text{id}}.$$

Theorem 4 (Deployment-error decomposition). *Let G be an event on which every stored planner is uniformly η -accurate for its own kernel. If $\Pr(G^c) \leq \alpha$ and the selected label is wrong with conditional probability at most β on G , then for every $r \in \mathcal{R}_1$,*

$$\mathbb{E}[\text{gap}(r)] \leq \eta + \alpha + \beta, \quad \Pr\left\{\sup_r \text{gap}(r) > \eta\right\} \leq \alpha + \beta.$$

Corollary 1 (Rates under SEG). *Assume $\text{SEG}(B, \mathbf{n}, \delta_{\text{seg}})$ and $n_m \geq N_{\text{stat}}(\eta, \delta_{\text{rfe}})$ for every eligible mode, where N_{stat} is the episode complexity of the invoked stationary RFE module under the same reward convention as (1). If deployment identification has error at most δ_{id} , then DME is $(\eta, \delta_{\text{seg}} + M_\star \delta_{\text{rfe}} + \delta_{\text{id}})$ -PAC. Invoking the exploration guarantee of Jin et al. [14] is valid only when accepted trajectories meet that paper’s sampling interface; detector-generated data do not inherit the rate automatically. Because $\mathcal{R}_1$ is contained in the class of per-step $[0, 1]$ rewards, any uniform guarantee for the larger class implies the same numerical η on $\mathcal{R}_1$.*

5.4 Diagnosis

Proposition 1 (Known-model maximum-likelihood diagnosis). *For a known finite family and fixed σ ,*

$$\Pr_m(\hat{m} \neq m) \leq \sum_{\ell \neq m} \text{Aff}(\mathbb{P}_m^\sigma, \mathbb{P}_\ell^\sigma).$$

Hence the worst-case error is at most $(M_\star - 1) \exp(-\min_{m \neq \ell} [-\log \text{Aff}])$.

Theorem 5 (Known-model diagnostic sample complexity). *Suppose two kernels differ by TV at least Δ at some triple $(h^\star, s^\star, a^\star)$, and a (possibly history-dependent) diagnostic policy visits that triple with probability at least μ in every episode. Then $q \geq (32/(\mu\Delta^2))(S \log 2 + \log(4\bar{M}/\delta_{\text{id}}))$ episodes suffice for a threshold test on the empirical next-state distribution to have error at most δ_{id} uniformly over at most $\bar{M}$ known candidates.*

Lemma 6 (Probe-stream two-window test). *Let observations take values in a finite alphabet of size d , and suppose a predictable probe produces i.i.d. samples with law $\nu^{(z_t)}$ while the mode is constant. Assume $\min_{m \neq \ell} \text{TV}(\nu^{(m)}, \nu^{(\ell)}) \geq \Delta$. For windows of length n , alarm when $\text{TV}(\hat{\nu}_{\text{left}}, \hat{\nu}_{\text{right}}) > \Delta/2$. If $n \geq (8/\Delta^2)(d + \log(4T/\delta))$, then over T candidate split times: (i) every pair of pure same-mode windows alarms with probability at most δ ; (ii) every pair of pure different-mode windows fails to alarm with probability at most δ . Delay is therefore $O(n)$ after a switch, provided both windows are pure.*

Lemma 6 is a building block for i.i.d. probe streams. MDP occupancy is not i.i.d. unless detection is restricted to a non-adaptive probe with episodic resets (or an explicit mixing argument). The next theorem makes that restriction and supplies matching, rollback quarantine, and quotas.

Theorem 6 (SEG for probe-window DME). *Assume 1–4. Let π_0 be a fixed non-adaptive probe policy such that one probe episode, started from ρ , yields an observation in a finite alphabet of size d whose law $\nu^{(m)}$ in mode m satisfies $\min_{m \neq \ell} \text{TV}(\nu^{(m)}, \nu^{(\ell)}) \geq \Delta$. Consider Algorithm 1 with the following modules, using probe episodes only for detection and matching: two-window TV test of Lemma 6 with*

$$n \geq \frac{8}{\Delta^2} (d + \log(8JT/\delta_{\text{seg}})),$$

where J is the number of blocks and T the number of candidate splits; rollback of the last n accepted probe episodes from the active library at an alarm; quarantine $B = n$ episodes on each side of the alarm; confirmation windows of length at least n ; Algorithm 2 with $\gamma = \Delta/2$; and Weissman TV balls of radius $\rho = \Delta/4$. If $D_{\min} \geq 2n + L_{\text{match}} + N_{\text{stat}}(\eta, \delta_{\text{rfe}})$, then the output satisfies $\text{SEG}(n, \mathbf{n}, \delta_{\text{seg}})$.

The short-dwell residual detector with a fixed TV floor 0.30 is a heuristic. Its windows are typically shorter than n , so Theorem 6 does not cover those runs.

Proposition 2 (Robust diagnosis under a TV margin). *Suppose each library set $\mathcal{C}_i$ is a TV ball of radius $\rho < \Delta/4$ about the true kernel of its label, and two modes differ by at least Δ at a probe triple visited with probability μ . Then the known-model test of Theorem 5 with threshold $\Delta/2 - \rho$ has the same sample complexity with Δ replaced by $\Delta - 2\rho$.*

5.5 Impossibility and obstructions

Theorem 7 (Zero-diagnostic impossibility). *Even if training reveals a two-mode library exactly, with $q = 0$ no algorithm can guarantee deployment loss below $1/2$ for all seen modes and all $r \in \mathcal{R}_1$.*

Theorem 8 (Finite-diagnostic testing lower bound). *For any adaptive σ , modes m, ℓ , and test ψ ,*

$$\Pr_m(\psi = \ell) + \Pr_\ell(\psi = m) \geq 1 - \text{TV}(\mathbb{P}_m^\sigma, \mathbb{P}_\ell^\sigma) \geq \frac{1}{2} \exp(-\mathcal{K}_q^\sigma(m, \ell)).$$

If selecting the wrong mode's policy loses at least γ in both directions, equal-prior expected deployment regret is at least $\frac{\gamma}{4} \exp(-\mathcal{K}_q^\sigma(m, \ell))$.

Theorem 9 (Occupancy obstruction). *Fix any algorithm and any $N_0 \geq 1$. There exist two-mode instances in which the second mode appears for N_0 training episodes and no (ε, δ) -PAC guarantee with $\varepsilon < 1/8$ and $\delta < 1/8$ can hold uniformly over $\mathcal{R}_1$ for deployment in that mode, whenever N_0 is smaller than a universal constant times $\varepsilon^{-2} \log(1/\delta)$. The construction is self-contained and does not multiply a published stationary lower bound by M .*

Theorem 10 (Behavioral unidentifiability). *If $P^{(m)}$ and $P^{(\ell)}$ induce the same trajectory law under every policy, then no reward-free procedure can distinguish them, and they induce identical values for every r . Conversely, if they differ on a reachable triple, some (possibly history-dependent) policy witnesses a positive one-step KL.*

5.6 Open targets

The following statements remain open.

Target 1 (direct sum). For an explicit product class of M independently hard, oracle-labelled modes, prove an adaptive-allocation lower bound of order $M N_{\text{stat}}^*(S, A, H, \varepsilon, \delta/M)$. Restricting to one mode recovers the stationary lower bound of Jin et al. [14]; multiplying that bound by M is not a proof.

Table 2: Tabular pilot means over 40 seed-configuration units. Lower gap is better. Generated from `results/recurrent_tabular_pilot/results.json`.

Method	Units	Accept rate	Mean worst-reward gap
Pooled	40	1.000000	0.134720962030
Restart, no reuse	40	1.000000	0.002482741316
Cluster, no quarantine	40	1.000000	0.001268908213
Recurrence-aware	40	1.000000	0.001378200021
Oracle boundary	40	1.000000	0.001627412643
Oracle mode	40	1.000000	0.001566496241
Sliding window	40	1.000000	0.154578714875

Target 2 (minimax switch overhead). Prove matching lower bounds on $D_{\min}$ and on the number of quarantined episodes as functions of $(\mu_{\text{probe}}, \Delta, S, A, H, \delta)$. Lemma 6 and Theorem 6 give an $O(n)$ delay upper bound, not a matching lower bound.

The short-window residual heuristic used in RFE-Recurrent-Bench is also not covered by Theorem 6.

6 Experiments

Tabular protocol. The retained finite-horizon protocol uses hash `6e54042d78e70be4`, seeds 0–9, nine states, mode counts $\{2, 3\}$, separations $\{0.75, 0.9\}$, dwell 500 transitions, three cycles, a 96-step deployment prefix, and horizon 20. Per-stage reward is at most 0.05, so pathwise reward is at most one. Crossing seeds, mode counts, and separations yields 40 units. Rewards, including adversarial placements, are revealed only after collection. Non-oracle methods use transition triples only.

We compare pooling, restart without reuse, clustering without quarantine, recurrence-aware DME, oracle-boundary and oracle-mode estimators, and a sliding window. The reported metric is the maximum value gap over a finite post-hoc reward suite. This finite maximum is *not* $\sup_{r \in \mathcal{R}_1}$.

Tabular findings. Pooling and sliding windows fail when action semantics conflict across modes (gaps 0.135 and 0.155). Recurrence-aware DME attains gap 0.00138, restart attains 0.00248, and clustering without quarantine attains 0.00127. Mean transition savings versus restart are 48.12 (199 of 200 recurrence rows positive). Deployment acceptance is 1.0 and learned mode count is correct in all 40 units. The paired restart-minus-DME improvement is 0.00110, with stored seed-bootstrap 95% interval $[0.00057, 0.00178]$. The interval excludes zero, but the mean is below the predeclared 0.01 practical margin, so the tabular gate *fails*. Clustering without quarantine is slightly better than DME on this suite; quarantine is not empirically justified here. That observation is compatible with Theorem 2: quarantine is a worst-case decontamination device, and this suite may contain few mixed windows relative to dwell. Theorem 1 explains the large pooled and sliding-window gaps: conflicting action semantics make the mixture kernel uniformly suboptimal.

RFE-Recurrent-Bench. Because the original synthetic suite saturates restart, we added a generative benchmark of four short-dwell tasks: a conflicting-action chain, RiverSwim with a reversed

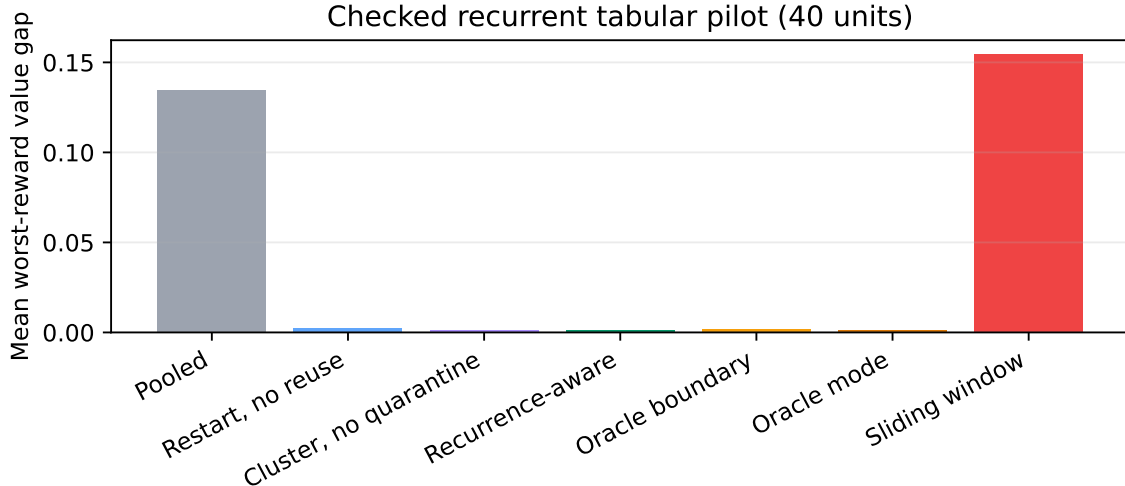


Figure 1: Tabular mean worst-reward value gap. Lower is better.

Table 3: RFE-Recurrent-Bench, ten seeds. Lower gap is better. Generated from `results/recurrent_bench_pilot/results.json`.

Task	Pooled	Restart	DME	Oracle mode
swap_chain	0.175	0.039	0.002	0.002
riverswim	0.515	0.275	0.164	0.076
deepsea	0.636	0.042	0.039	0.000
four_rooms	0.005	0.215	0.076	0.025

current, a DeepSea-style dive-action swap, and Four Rooms with moving doors. Collection remains reward-free. Hash `90f18f4bd17c4218`, ten seeds, 40 units (Table 3 and Figure 2).

The implemented detector originally almost never fired, so DME collapsed to pooling. After correcting the two-window threshold, cooldown, and adding an active-mode residual test, recurrence-aware DME attains mean gap 0.070 versus 0.143 for restart, 0.333 for pooling, and 0.026 for oracle-mode. The paired restart-minus-DME improvement is 0.073 (bootstrap 95% interval $[0.040, 0.109]$), above the predeclared 0.01 margin, so the benchmark gate *passes*. Swap-chain and DeepSea are near the oracle. RiverSwim still pays a quarantine cost relative to clustering without quarantine (0.164 versus 0.094). The implementation now rolls back the pre-alarm tail so mixed counts do not remain in the library; those runs are a new protocol and do not overwrite hash `90f18f4bd17c4218`. Tabular Four Rooms remains easy for pooling (gap 0.005). Collection uses a greedy covering collector in the covering-then-plan template of Jin et al. [14]. A likelihood-matching baseline (`mbcd_like`) is included in the tabular harness as a reward-free analogue of context detection, not as the published MBCD algorithm.

MiniGrid stress test. A ten-seed pilot uses symbolic 7×7 partial observations and two deterministic recurring transition/observation regimes (Table 4). Recurrence-aware learning has worst-task gap 0.0492 versus 0.2648 for restart, saves 6.8 samples, and has deployment ID error 0.20. Pooling has zero measured gap, so the scale gate fails. The oracle is an empirical label-aware control, not an exact optimum. Observations are symbolic, not RGB.

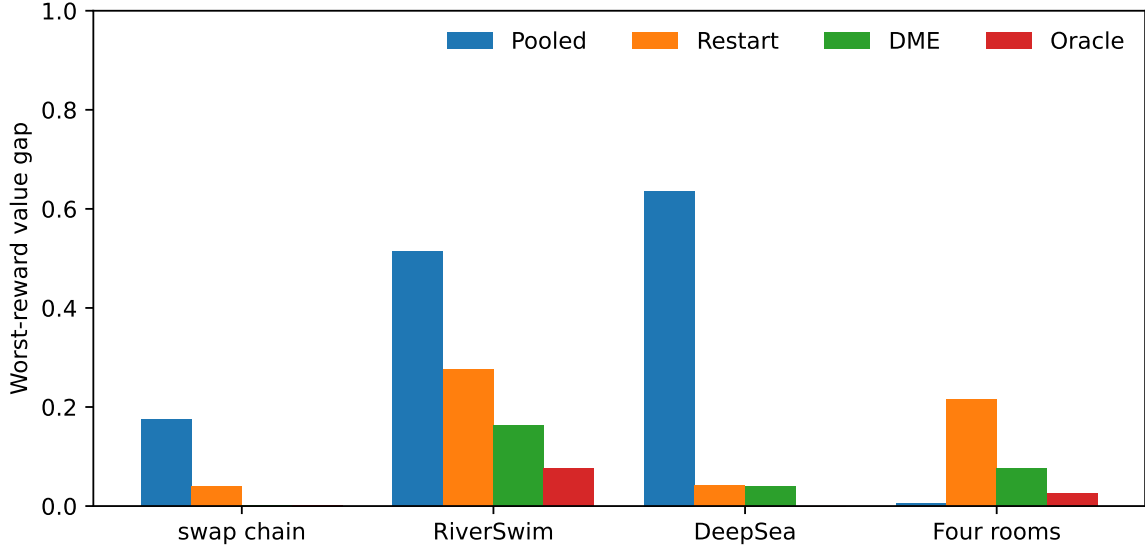


Figure 2: Per-task worst-reward gaps on RFE-Recurrent-Bench.

Table 4: MiniGrid pilot means over ten seeds.

Method	Seeds	Mean worst-task gap
pooled	10	0.000000
restart	10	0.264844
recurrence_aware	10	0.049219
oracle_upper_bound	10	0.000000

7 Limitations

The modular PAC theorem plus Theorem 6 give an end-to-end guarantee only for probe-window DME with the stated window length. The short-window residual heuristic and the greedy covering collector are not that guarantee: covering-then-plan follows Jin et al. [14] only when the covering certificate meets their occupancy condition. Empirical rewards are a finite suite. MiniGrid is too easy for pooling. Strong separability and dwell assumptions exclude gradual drift and reward-only changes. Wrong diagnosis can select a harmful controller; certified abstention is required for any future deployment.

8 Conclusion

Regime-conditional RFE is a distinct objective at the intersection of reward-free learning, recurring hidden dynamics, and active identification. DME makes the module boundaries explicit. The proved results isolate what follows from valid segmentation and what is information-theoretically impossible without diagnosis. Probe-window DME satisfies SEG under an explicit window length (Theorem 6); the short-window residual heuristic used in the passing benchmark is not that theorem. The original saturated tabular and MiniGrid gates fail. An M -fold direct-sum lower bound remains open.

A Notation

Symbol	Meaning
S, A, H	state, action cardinalities and horizon
$P^{(m)}$	episodic kernel of mode m
$M_\star \leq \bar{M}$	unknown and known mode counts
z_k, Z	hidden training and deployment labels
q	diagnostic episode budget
$\mathcal{R}_1$	rewards obeying (1)
SEG	segmentation contract (Definition 2)
$\text{TV}, D_{\text{KL}}, \text{Aff}$	total variation, KL, Hellinger affinity
$\mathcal{B}_q, \mathcal{K}_q^\sigma$	controlled affinity and KL coefficients
$D_{\min}, B$	minimum dwell and quarantine length

B Proofs

B.1 Proof of Lemma 1

Let $\tau = (s_1, a_1, \dots, s_H, a_H)$ and write $R(\tau) = \sum_{h=1}^H r_h(s_h, a_h)$. Condition (1) yields $R(\tau) \in [0, 1]$. Let μ and $\hat{\mu}$ be the laws of τ under (P, π) and $(\hat{P}, \pi)$. These laws factor as

$$\mu(\tau) = \rho(s_1) \prod_{h=1}^H \pi_h(a_h \mid s_h, h, \text{hist}) P_h(s_{h+1} \mid s_h, a_h),$$

and likewise for $\hat{\mu}$, with the same ρ and π . Therefore the policies cancel in the likelihood ratio. A standard telescoping argument for trajectory laws [13] produces a coupling under which

$$\text{TV}(\mu, \hat{\mu}) \leq \sum_{h=1}^H \mathbb{E}_{\hat{\mu}}[\text{TV}(P_h(\cdot \mid s_h, a_h), \hat{P}_h(\cdot \mid s_h, a_h))].$$

On the support of π each summand is at most ξ , so $\text{TV}(\mu, \hat{\mu}) \leq H\xi$. For any $[0, 1]$ -valued random variable, $|\mathbb{E}_\mu R - \mathbb{E}_{\hat{\mu}} R| \leq \text{TV}(\mu, \hat{\mu})$, which is the value bound.

For the planning claim, let $\hat{\pi}$ be optimal in $\hat{P}$. Then

$$V_P^\star(r) - V_{\hat{P}}^{\hat{\pi}}(r) \leq |V_P^\star(r) - V_{\hat{P}}^\star(r)| + (V_{\hat{P}}^\star(r) - V_{\hat{P}}^{\hat{\pi}}(r)) + |V_{\hat{P}}^{\hat{\pi}}(r) - V_P^{\hat{\pi}}(r)|.$$

The middle term is zero. The first term is at most $\sup_\pi |V_P^\pi(r) - V_{\hat{P}}^\pi(r)| \leq H\xi$ by the value bound, and the last term is at most $H\xi$ likewise.

B.2 Proof of Lemma 2

Let X be the diagnostic transcript $(s_{t,h}, a_{t,h}, s_{t,h+1})_{t \leq q, h \leq H}$. The strategy σ selects $a_{t,h}$ as a (randomized) function of the strict past $H_{t,h}$. The density of X under mode m , with respect to a dominating product measure, therefore factors as

$$p_m(X) = \prod_{t=1}^q \prod_{h=1}^H \sigma(a_{t,h} \mid H_{t,h}) P_h^{(m)}(s_{t,h+1} \mid s_{t,h}, a_{t,h}),$$

and likewise for ℓ . The factors σ cancel in p_m/p_ℓ . The chain rule for KL [7, Theorem 2.5.3] yields

$$D_{\text{KL}}(\mathbb{P}_m^\sigma \| \mathbb{P}_\ell^\sigma) = \sum_{t,h} \mathbb{E}_m \left[D_{\text{KL}}(P_h^{(m)}(\cdot | s_{t,h}, a_{t,h}) \| P_h^{(\ell)}(\cdot | s_{t,h}, a_{t,h})) \right].$$

If a triple is never visited, the corresponding conditional KL is never evaluated and contributes zero to the expectation. If it is visited and $P_h^{(\ell)}(\cdot | s, a)$ fails to dominate $P_h^{(m)}(\cdot | s, a)$, the conditional KL is $+\infty$.

B.3 Proof of Lemma 3

Pinsker's inequality $2 \text{TV}^2 \leq D_{\text{KL}}$ is standard [25, Lemma 2.5]. Le Cam's inequality $\text{TV} \leq \sqrt{1 - \text{Aff}^2}$ follows from writing

$$\text{TV}(P, Q) = \frac{1}{2} \int |p - q| = \frac{1}{2} \int |\sqrt{p} - \sqrt{q}| |\sqrt{p} + \sqrt{q}|$$

and Cauchy-Schwarz:

$$\left(\frac{1}{2} \int |\sqrt{p} - \sqrt{q}| |\sqrt{p} + \sqrt{q}| \right)^2 \leq \left(\frac{1}{2} \int (\sqrt{p} - \sqrt{q})^2 \right) \left(\frac{1}{2} \int (\sqrt{p} + \sqrt{q})^2 \right) = (1 - \text{Aff})(1 + \text{Aff}) = 1 - \text{Aff}^2.$$

Rearrangement gives $\text{Aff} \leq \sqrt{1 - \text{TV}^2}$. The elementary bound $1 - u \leq e^{-u}$ with $u = \text{TV}^2$ yields $\sqrt{1 - \text{TV}^2} \leq e^{-\text{TV}^2/2}$.

B.4 Proof of Theorem 1

Coordinatewise, $M_h(\cdot | s, a) - P_h(\cdot | s, a) = (1 - \alpha)(Q_h(\cdot | s, a) - P_h(\cdot | s, a))$, so taking $\frac{1}{2} \|\cdot\|_1$ proves the TV identity.

For inconsistency, reuse the instance of Theorem 7: one-step MDP, actions $\{a_0, a_1\}$, terminals $\{g, b\}$, mode 0 sending a_0 to g and a_1 to b , mode 1 reversed, and $r(g) = 1$. The mixture $M = \frac{1}{2}P^{(0)} + \frac{1}{2}P^{(1)}$ sends both actions to g with probability $1/2$. Every policy is optimal for M and has value $1/2$ in both true modes, while $V^* = 1$. Infinite mixed data recover M exactly and still yield gap $1/2$.

B.5 Proof of Theorem 2

Condition on the n mode labels in the window and on the (predictable) policies. Each accepted transition at a fixed (h, s, a) is drawn from P_h or Q_h according to the episode's mode. The empirical next-state frequency is the corresponding mixture of empirical measures, whose conditional expectation is the displayed convex combination. Taking $\text{TV}(\cdot, P_h)$ and using Theorem 1 with $\alpha = n_0/n$ gives the bias identity. If every episode whose window straddles a true boundary lies in the B episodes at each block end, those episodes are excluded from accepted counts and the identity is never applied to a mixed window.

B.6 Proof of Lemma 4

Write $d_\infty(P, Q) = \max_{h,s,a} \text{TV}(P_h, Q_h)$. By assumption $d_\infty(P^{(m)}, P^{(\ell)}) \geq \Delta$ for $m \neq \ell$. A TV-ball of radius $\rho < \Delta/4$ around $P^{(m)}$ is disjoint from the ball of radius ρ around $P^{(\ell)}$, and the gap between balls is greater than $\Delta/2$. Hence if $\mathcal{C}_{\text{new}}$ and $\mathcal{C}_i$ are balls of radius ρ around the same kernel, $d_i = 0$; if they wrap distinct kernels, $d_i > \Delta/2 = \gamma$. Algorithm 2 therefore returns the unique true index on a recurrence and NEW on an unseen separated mode. If radii fail to be valid, the lemma does not apply.

B.7 Proof of Lemma 5

Under SEG, each of the J blocks contributes at most $2B$ quarantined episodes and therefore at least $D_{\min} - 2B$ accepted episodes, all assigned to the same library index. Restart that keeps only the last block has at most $D_{\min}$ episodes (even without quarantine). The comparison $D_{\min} < N \leq J(D_{\min} - 2B)$ is arithmetic.

B.8 Proof of Theorem 3

Let E_{seg} be the SEG event, E_m the success of the stationary RFE call for mode m , and E_{id} correct deployment identification. The hypotheses are $\Pr(E_{\text{seg}}^c) \leq \delta_{\text{seg}}$, $\Pr(E_m^c \mid E_{\text{seg}}) \leq \delta_{\text{rfe}}$, and $\Pr(E_{\text{id}}^c \mid E_{\text{seg}} \cap \bigcap_m E_m) \leq \delta_{\text{id}}$. A union bound gives

$$\Pr\left(E_{\text{seg}} \cap \bigcap_{m=1}^{M_\star} E_m \cap E_{\text{id}}\right) \geq 1 - \delta_{\text{seg}} - M_\star \delta_{\text{rfe}} - \delta_{\text{id}}.$$

On this intersection, Definition 2 supplies a bijection from library indices to true modes, every accepted sample is pure, each planner Π_i is uniformly η -accurate for its kernel, and $\hat{m}$ equals the library index of Z . Therefore $\sup_{r \in \mathcal{R}_1} \text{gap}(r) \leq \eta$.

B.9 Proof of Theorem 4

On $G \cap \{\hat{m} = Z\}$ the gap is at most η uniformly in r . On the complement the gap is at most 1 by (1). Hence

$$\mathbb{E}[\text{gap}(r)] \leq \eta \Pr(G \cap \{\hat{m} = Z\}) + 1 \cdot \Pr(G^c \cup \{\hat{m} \neq Z\}).$$

The event $G^c \cup \{\hat{m} \neq Z\}$ is contained in $G^c \cup (G \cap \{\hat{m} \neq Z\})$, which has probability at most $\alpha + \beta$. The same inclusion yields the high-probability bound because on $G \cap \{\hat{m} = Z\}$ one has $\sup_r \text{gap}(r) \leq \eta$.

B.10 Proof of Corollary 1

Apply Theorem 3 with the given quotas. The inclusion $\mathcal{R}_1 \subset \{r : r_h \in [0, 1]\}$ is immediate from (1), so any guarantee stated for the larger class remains valid on $\mathcal{R}_1$ at the same η . The warning about sampling interfaces is the content of Definition 2(iv): purity does not imply that a published RFE theorem applies.

B.11 Proof of Proposition 1

Let p_j be densities of $\mathbb{P}_j^\sigma$ under a common dominating measure. Maximum likelihood errs in favour of ℓ only on $\{p_\ell \geq p_m\}$, and

$$\Pr_m(p_\ell \geq p_m) = \int_{\{p_\ell \geq p_m\}} p_m \leq \int \sqrt{p_m p_\ell} = \text{Aff}(\mathbb{P}_m^\sigma, \mathbb{P}_\ell^\sigma),$$

because $p_m \leq \sqrt{p_m p_\ell}$ on that set. A union bound over $\ell \neq m$ proves the claim. Writing $\text{Aff} = \exp(\log \text{Aff})$ gives the exponential form.

B.12 Proof of Theorem 5

Fix a distinguishing triple (h^*, s^*, a^*) at which two candidates differ by TV at least Δ . Let N be the number of diagnostic visits to that triple over q episodes. Each episode contributes a visit with probability at least μ , so N stochastically dominates $\text{Bin}(q, \mu)$. A Chernoff bound yields $\Pr(N < \mu q/2) \leq \exp(-\mu q/8)$.

On the event $\{N \geq \mu q/2\}$, let $\hat{p}$ be the empirical next-state distribution at the triple. The test returns the unique library kernel whose TV distance to $\hat{p}$ is at most $\Delta/4$, and rejects if none or several exist. If the truth $P^{(m)}$ satisfies $\text{TV}(\hat{p}, P^{(m)}) \leq \Delta/4$, then every competitor $P^{(\ell)}$ has $\text{TV}(\hat{p}, P^{(\ell)}) \geq \Delta - \Delta/4 = 3\Delta/4$, so the test is correct.

Weissman's inequality [27] gives, for an alphabet of size S ,

$$\Pr(\text{TV}(\hat{p}, P^{(m)}) > \Delta/4 \mid N = n) \leq (2^S - 2) \exp(-2n(\Delta/4)^2) = (2^S - 2) \exp(-n\Delta^2/8).$$

Choosing $q \geq (32/(\mu\Delta^2))(S \log 2 + \log(4\bar{M}/\delta_{\text{id}}))$ makes $\mu q/2 \geq (16/\Delta^2)(S \log 2 + \log(4\bar{M}/\delta_{\text{id}}))$, hence

$$\exp(-\mu q/8) \leq \frac{\delta_{\text{id}}}{4\bar{M}}, \quad (2^S) \exp(-(\mu q/2)\Delta^2/8) \leq \frac{\delta_{\text{id}}}{4\bar{M}}.$$

A union bound over at most $\bar{M}$ candidates completes the proof.

B.13 Proof of Lemma 6

Let ν be the common law of a pure same-mode window of length n . Then

$$\text{TV}(\hat{\nu}_{\text{L}}, \hat{\nu}_{\text{R}}) \leq \text{TV}(\hat{\nu}_{\text{L}}, \nu) + \text{TV}(\hat{\nu}_{\text{R}}, \nu).$$

A false alarm requires the left-hand side to exceed $\Delta/2$, hence some window to have TV error greater than $\Delta/4$. Weissman's inequality yields

$$\Pr(\text{TV}(\hat{\nu}, \nu) > \Delta/4) \leq (2^d - 2) \exp(-n\Delta^2/8).$$

For two windows and T split times, the union bound is at most $2T(2^d) \exp(-n\Delta^2/8)$. The stated n makes this at most δ .

If the two windows are pure and drawn from laws at TV distance at least Δ , the same decomposition gives $\text{TV}(\hat{\nu}_{\text{L}}, \hat{\nu}_{\text{R}}) \geq \Delta - \text{TV}(\hat{\nu}_{\text{L}}, \nu^{(m)}) - \text{TV}(\hat{\nu}_{\text{R}}, \nu^{(\ell)})$. Both empirical errors being at most $\Delta/4$ produces an alarm. The failure probability is therefore controlled by the same estimate.

B.14 Proof of Theorem 6

Write $\delta = \delta_{\text{seg}}$. Apply Lemma 6 to the probe-episode stream with failure probability $\delta/4$ and T candidate splits. On that event: (i) no pair of pure same-mode windows of length n alarms; (ii) every pair of pure different-mode windows of length n alarms, so after a switch the delay is at most n once both windows are pure.

Rollback removes the last n accepted probe episodes from the active library at the alarm. Quarantine then discards n post-alarm episodes. Hence mixed windows are contained in the last n episodes of the old block and the first n of the new block, which is localization with $B = n$. Accepted interiors are therefore pure, which is (i) in Definition 2.

Confirmation windows of length at least n yield empirical laws within TV distance $\rho = \Delta/4$ of the true probe law, again by Weissman and a union bound of $\delta/4$ over at most J segments. Lemma 4 with $\gamma = \Delta/2$ therefore returns a unique recurrence or NEW, which is (iii). Weissman balls of radius

ρ about the pooled accepted probe counts cover the true kernels jointly with probability $1 - \delta/4$ over at most $\bar{M}d$ coordinates, which is (v).

Assumption 4 with $B = n$ yields at least $D_{\min} - 2n$ accepted episodes per block. Recurrence pooling and Assumption 3 give the RFE quota (iv). A union bound of four $\delta/4$ events completes SEG($n, \mathbf{n}, \delta$).

The argument uses i.i.d. probe episodes. It does not apply to a residual detector run on an adaptive RFE policy in a continuing MDP, nor to a TV threshold smaller than $\Delta/2$ with n below the displayed length.

B.15 Proof of Proposition 2

On the event that each $\mathcal{C}_i$ is a TV ball of radius ρ about the truth, any empirical diagnostic law $\hat{\nu}$ with $\text{TV}(\hat{\nu}, P^{(m)}) \leq (\Delta - 2\rho)/4$ lies at distance at most $\Delta/2 - \rho$ from the true ball and at least $3\Delta/4 - \rho$ from every competitor ball. The threshold $\Delta/2 - \rho$ therefore separates them. Theorem 5 applies with $\Delta \leftarrow \Delta - 2\rho$.

B.16 Proof of Theorem 7

Consider a one-step problem with initial state x , actions $\{a_0, a_1\}$, and terminals $\{g, b\}$. In mode 0, a_0 leads to g and a_1 to b ; in mode 1 the transitions are reversed. Let r assign 1 at g and 0 elsewhere, which satisfies (1). With $q = 0$ the algorithm must choose a (possibly randomized) action at x independently of the deployment label. If it selects a_0 with probability p , the losses are $1 - p$ and p . Their maximum is at least $1/2$. Both modes may have appeared arbitrarily often during training; the missing information is which one is deployed.

B.17 Proof of Theorem 8

For binary hypotheses, $\inf_{\psi} (\Pr_m(\psi = \ell) + \Pr_\ell(\psi = m)) = 1 - \text{TV}(\mathbb{P}_m^\sigma, \mathbb{P}_\ell^\sigma)$ [17]. Bretagnolle–Huber [3, 5] gives $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$. Set $x = e^{-D_{\text{KL}}} \in [0, 1]$. The inequality $1 - \sqrt{1 - x} \geq x/2$ holds because $f(x) = 1 - \sqrt{1 - x} - x/2$ satisfies $f(0) = 0$ and $f'(x) = \frac{1}{2}(1 - x)^{-1/2} - 1/2 \geq 0$ on $[0, 1]$. Therefore $1 - \text{TV} \geq \frac{1}{2}e^{-D_{\text{KL}}}$. Under equal priors the error probability is at most half of the two-hypothesis sum and at least half of its infimum; multiplying the infimum by $\gamma/2$ (the equal-prior mixture of the two directional losses) yields the regret bound $\frac{\gamma}{4} \exp(-D_{\text{KL}})$.

B.18 Proof of Theorem 9

We give a self-contained two-action construction. Let $H = 1$, with initial state x , actions $\{a_+, a_-\}$, and terminals $\{g, b\}$. Mode 0 is deterministic: both actions lead to g . Mode $\theta \in \{+\gamma, -\gamma\}$ has

$$P(g \mid x, a_+) = \frac{1}{2} + \theta, \quad P(g \mid x, a_-) = \frac{1}{2} - \theta,$$

with $0 < \gamma \leq 1/4$. Let r assign 1 at g and 0 at b , which lies in $\mathcal{R}_1$. Then $V^* = 1/2 + \gamma$ and the suboptimal action has gap 2γ .

Take $\gamma = \varepsilon$ with $\varepsilon < 1/8$, so a wrong action has gap $2\varepsilon > \varepsilon$. The common mode is uninformative about θ . All information about the rare mode is contained in its N_0 episodes. Each episode, under any action, yields a Bernoulli observation whose KL between $+\gamma$ and $-\gamma$ is at most $\chi^2 = 4\gamma^2/(\frac{1}{4} - \gamma^2) \leq 32\gamma^2$ for $\gamma \leq 1/4$. Thus $\mathcal{K}_{N_0} \leq 32N_0\varepsilon^2$. Theorem 8 yields two-hypothesis error at least $\frac{1}{2} \exp(-32N_0\varepsilon^2)$. If $N_0 \leq (1/32)\varepsilon^{-2} \log 2$, this is at least $1/4$. Any algorithm that then selects an action for the rare mode therefore has $\Pr(\text{gap} > \varepsilon) \geq 1/4$, so it cannot be (ε, δ) -PAC for $\delta < 1/8$.

B.19 Proof of Theorem 10

If two kernels induce the same law under every policy, the observed training and diagnostic transcripts are identically distributed, so any test has error at least $1/2$ in the two-hypothesis problem. Values coincide because they are expectations of (1) under those laws. Conversely, if $P_h^{(m)}(\cdot | s, a) \neq P_h^{(\ell)}(\cdot | s, a)$ and some policy reaches (s, a) at step h with positive probability from ρ , then the one-step KL at that triple is strictly positive (or infinite). Lemma 2 therefore yields $\mathcal{K}_q^\sigma(m, \ell) > 0$ for a strategy that targets the triple, hence $\mathcal{B}_q > 0$.

C Reproducibility

From the repository root:

```
python3 -m pytest -q
python3 run.py --protocol recurrent-tabular --profile pilot \
  --output results/recurrent_tabular_pilot
python3 run.py --protocol minigrid-recurrent --profile pilot \
  --output results/minigrid_recurrent_pilot
python3 run.py --protocol recurrent-bench --profile pilot \
  --output results/recurrent_bench_pilot
python3 scripts/generate_regime_rfe_assets.py
python3 scripts/generate_benchmark_assets.py
tectonic paper/regime_rfe.tex --keep-logs
```

Tables and figures are generated from retained JSON/CSV. The official NeurIPS style file was unavailable locally; the compiled PDF uses the declared article fallback.

References

- [1] Lucas N. Alegre, Ana L. C. Bazzan, and Bruno C. da Silva. Minimum-delay adaptation in non-stationary reinforcement learning via online high-confidence change-point detection. In *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems*, pages 97–105. IFAAMAS, 2021. URL <https://www.ifaamas.org/Proceedings/aamas2021/pdfs/p97.pdf>.
- [2] Mohammad Gheshlaghi Azar, Ian Osband, and Rémi Munos. Minimax regret bounds for reinforcement learning. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, pages 263–272. PMLR, 2017.
- [3] Jean Bretagnolle and Catherine Huber. Estimation des densités: risque minimax. In *Séminaire de Probabilités XII*, pages 342–363. Springer, 1978. doi: 10.1007/BFb0064610. URL https://www.numdam.org/item/SPS_1978__12__342_0/.
- [4] Emma Brunskill and Lihong Li. Sample complexity of multi-task reinforcement learning. *arXiv preprint arXiv:1308.3513*, 2013. URL <https://arxiv.org/abs/1308.3513>.
- [5] Clément L. Canonne. A short note on an inequality between KL and TV. *arXiv preprint arXiv:2202.07198*, 2023. URL <https://arxiv.org/abs/2202.07198>.

- [6] Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Reinforcement learning for non-stationary MDPs: The blessing of (more) optimism. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119, pages 1843–1854. PMLR, 2020. URL <https://proceedings.mlr.press/v119/cheung20a.html>.
- [7] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2 edition, 2006.
- [8] Christoph Dann, Tor Lattimore, and Emma Brunskill. Unifying PAC and regret: Uniform PAC bounds for Markov decision processes. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [9] Xiaoming Duan, Yagiz Savas, Rui Yan, Zhe Xu, and Ufuk Topcu. On the detection of Markov decision processes. *arXiv preprint arXiv:2112.12338*, 2021. URL <https://arxiv.org/abs/2112.12338>.
- [10] Argyrios Gerogiannis, Yu-Han Huang, and Venugopal V. Veeravalli. DARLING: Detection augmented reinforcement learning with non-stationary guarantees. *arXiv preprint arXiv:2604.16684*, 2026. URL <https://arxiv.org/abs/2604.16684>.
- [11] Assaf Hallak, Dotan Di Castro, and Shie Mannor. Contextual MDPs. *arXiv preprint arXiv:1502.02259*, 2015. URL <https://arxiv.org/abs/1502.02259>.
- [12] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, non-parametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991. URL <https://arxiv.org/abs/1810.08240>.
- [13] Thomas Jaksch, Ronald Ortner, and Peter Auer. Near-optimal regret bounds for reinforcement learning. *Journal of Machine Learning Research*, 11:1563–1600, 2010. URL <https://jmlr.org/papers/v11/jaksch10a.html>.
- [14] Chi Jin, Akshay Krishnamurthy, Max Simchowitz, and Tiancheng Yu. Reward-free exploration for reinforcement learning. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119, pages 4870–4879. PMLR, 2020. URL <https://proceedings.mlr.press/v119/jin20d.html>.
- [15] Emilie Kaufmann, Pierre Ménard, Omar Darwiche Domingues, Anders Jonsson, Edouard Leurent, and Michal Valko. Adaptive reward-free exploration. In *Proceedings of the 32nd International Conference on Algorithmic Learning Theory*, volume 132, pages 517–539. PMLR, 2021. URL <https://proceedings.mlr.press/v132/kaufmann21a.html>.
- [16] Jeongyeol Kwon, Yonathan Efroni, Constantine Caramanis, and Shie Mannor. RL for latent MDPs: Regret guarantees and a lower bound. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL <https://proceedings.neurips.cc/paper/2021/hash/cd755a6c6b699f3262bcc2aa46ab507e-Abstract.html>.
- [17] Lucien Le Cam. Convergence of estimates under dimensionality restrictions. *The Annals of Statistics*, 1(1):38–53, 1973. doi: 10.1214/aos/1176342376.
- [18] Gary Lorden. Procedures for reacting to a change in distribution. *The Annals of Mathematical Statistics*, 42(6):1897–1908, 1971.

- [19] Pierre Ménard, Omar Darwiche Domingues, Anders Jonsson, Emilie Kaufmann, Edouard Leurent, and Michal Valko. Fast active learning for pure exploration in MDPs. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 7599–7608. PMLR, 2021. URL <https://proceedings.mlr.press/v139/menard21a.html>.
- [20] Dipendra Misra, Mikael Henaff, Akshay Krishnamurthy, and John Langford. Kinematic state abstraction and provably efficient rich-observation reinforcement learning. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119, pages 6961–6970. PMLR, 2020.
- [21] Ronald Ortner, Pratik Gajane, and Peter Auer. Variational regret bounds for reinforcement learning. In *Proceedings of the 35th Conference on Uncertainty in Artificial Intelligence*, volume 115, pages 81–90. PMLR, 2020.
- [22] Sindhu Padakandla. A survey of reinforcement learning algorithms for dynamically varying environments. *ACM Computing Surveys*, 54(6):1–35, 2021.
- [23] E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1–2):100–115, 1954.
- [24] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.
- [25] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.
- [26] Chen-Yu Wei and Haipeng Luo. Non-stationary reinforcement learning without prior knowledge: An optimal black-box approach. In *Proceedings of the 34th Conference on Learning Theory*, volume 134, pages 4300–4346. PMLR, 2021. URL <https://proceedings.mlr.press/v134/wei21b.html>.
- [27] Tsachy Weissman, Erik Ordentlich, Gadiel Seroussi, Sergio Verdú, and Marcelo J. Weinberger. Inequalities for the L_1 deviation of the empirical distribution. Technical report, Hewlett-Packard Laboratories, 2003.
- [28] Zihan Zhang, Simon S. Du, and Xiangyang Ji. Nearly minimax optimal reward-free reinforcement learning. *arXiv preprint arXiv:2010.05901*, 2021. URL <https://arxiv.org/abs/2010.05901>.